

Question number	Answer	Mark
7(a)(i)	The only correct answer is B <i>A is incorrect because Q is a hydrogen bond</i> <i>C is incorrect because R is a covalent bond</i> <i>D is incorrect because Q is a hydrogen bond</i>	(1)

Question number	Answer	Mark
7(a)(ii)	The only correct answer is C <i>A is incorrect because a deoxyribose does not have a negative charge</i> <i>B is incorrect because a covalent bond does not have a charge</i> <i>D is incorrect because the bases do not have a negative charge</i>	(1)

Question number	Answer	Additional guidance	Mark
7(b)(i)	{Maleo / this bird / it} is only found in {Indonesia / this country / forest}	DO NOT ACCEPT answer given in context of habitat	(1)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	the newly hatched birds cannot fly to the forest / the birds cannot get to beach	ACCEPT not enough beach left to lay eggs / not enough forest left for {roosting / feeding} / humans damage {eggs / nests} / decrease gene flow / inbreeding	(1)
Question number	Answer	Additional guidance	Mark
7(b)(iii)	An explanation that includes the following points: - because laying eggs in the sand will {keep them warm / hide them from predators / protect from wave action} (1) - because if the adults leave the site {there will be no indication that there are eggs laid there / they can go and find food (for themselves) / can avoid predators / it is too hot on the beach} (1) - because if the young can fly (straightaway) they can {avoid predators / get food} (1)		(3)

Question number	Answer	Additional guidance	Mark
7(b)(iv)	An explanation that includes the following points: - primers needed so that (DNA / taq) polymerase can bind (to one strand) (1) - because one primer is needed for each strand (1) - because the DNA strands {are antiparallel / are complementary / have different base sequences at each end} (1)	ACCEPT one primer can bind to each end of the DNA molecule ACCEPT two primers each with a complementary sequence of bases are needed DO NOT ACCEPT one for 3' and one for 5' end	(2)

Question number	Answer	Additional guidance	Mark
7(b)(v)	An explanation that includes the following points: • this sample of eggs had more males than females so the {birds may not be monogamous / claim may not be true} (1) • the sample size is too small (1) • the number of chicks may not represent the number of adults (1)	ACCEPT fewer females than males any correct <u>manipulated</u> data that illustrates this, including not a 1:1 ACCEPT a suitable description of why e.g. more males die	(3)

